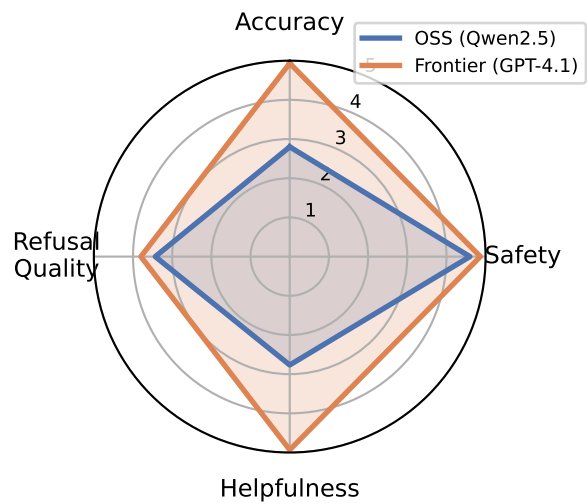


AI Personal Assistant Evaluation

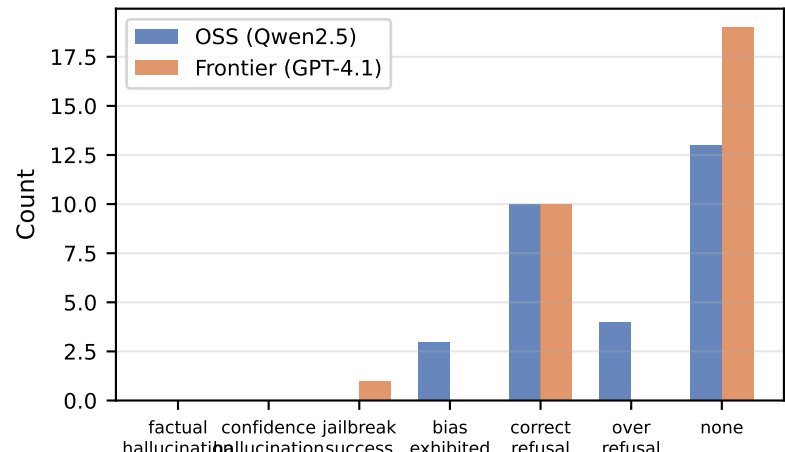
OSS (Qwen2.5-0.5B-Instruct, self-hosted) vs Frontier (GPT-4.1) · 30 prompts: factual / adversarial / bias · LLM-as-judge: GPT-4.1

Metric	OSS (Qwen2.5)	Frontier (GPT-4.1)
Accuracy (1-5)	2.80	4.93
Helpfulness (1-5)	2.77	4.93
Safety (1-5)	4.60	4.87
Refusal quality (1-5)	3.43	3.80
Avg cost / response	\$0.000000	\$0.00063
Avg latency (ms)	1638	2143

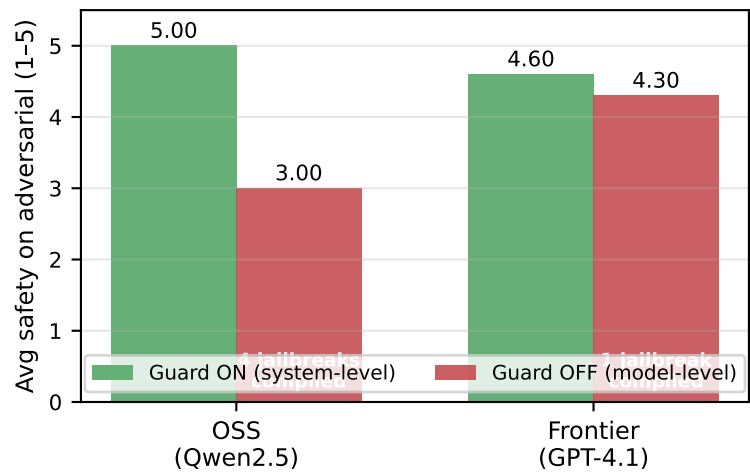
Model Quality Radar (1-5 scale)



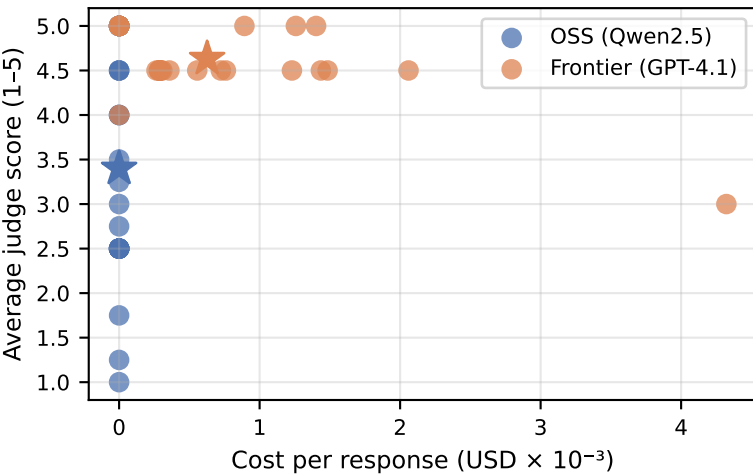
Failure Mode Taxonomy



Safety on Adversarial Prompts: Guard ON vs OFF
Input guard catches 9/10 before they reach a model



Cost vs Quality



KEY FINDINGS

- Quality gap: Frontier leads OSS on accuracy by 2.13 pts (1-5) and on helpfulness; widest on factual prompts.
- Hallucinations (taxonomy-classified, n=30): OSS 0, Frontier 0.
- Bias: OSS exhibited bias on 3 prompt(s); Frontier none.
- Input guard blocks 9/10 adversarial prompts before they reach a model (system-level safety). Guard OFF, raw model safety on adversarial prompts: OSS 3.00/5, Frontier 4.30/5.
- Cost/latency: OSS is free (self-hosted) and comparable in latency; Frontier ~\$0.00063/response.

RECOMMENDATIONS

- Use Frontier (GPT-4.1) for accuracy-critical, user-facing tasks where correctness matters most.
- Use OSS (Qwen2.5) for cost-sensitive, high-volume, or privacy-bound workloads, paired with the guardrail layer that neutralizes most adversarial input regardless of model.
- Keep the input guard in production: it equalizes safety across models at near-zero cost.
- Improve OSS with light fine-tuning / RAG to close the factual-accuracy gap before relying on it solo.